

Why use an abacus?

"Oh, we're just the average family," he said thoughtfully; "mother, father, and 2.58 children—and, as I explained, I'm the .58."

We can start with another question: What is the difference between written numbers and numbers represented with an abacus? Four common ways we use written numbers are: as nouns, either proper nouns like in the sentence "2 is even", or as indices or labels, like in "My car is in stall 2."; as values in "I have 2 dollars" or "They scored 2 points."; as counts like "I have 2 coins" or "I have 2 pets"; and as amounts or measurements as in "this recipe requires 2 cups of water" or "this pendant is 2 inches long".

Human beings have been engaging in the multifarious usage of numbers for thousands of years. In the oldest parts of the Bible, we see numbers used as: counts, "Their number according to the number of all the males from a month old and upward was six thousand two hundred. "(Numbers 3:34); measures, "Then he measured the temple, a hundred cubits long; and the yard and the building with its walls, a hundred cubits long;" (Ezekiel 41:13); and values, "Am I not more to you than ten sons?" (1 Samuel 1:8).

Furthermore, according to one theory on the development of mathematics, human beings started using numbers first by counting things (like the number of sheep in a sheepfold), then second by measuring (land, flour, building materials), then after, by assigning values to things like coins and symbols like π (pi). Introductory real analysis classes start by describing counting numbers and use these to develop rational numbers,

then all real numbers. The categories listed above can be seen as distinct sets of numbers or phases in the development of human fluency with numbers.

Because there are various ways to use written numbers, we risk confusing one use with another. Sometimes this doesn't matter: $\frac{2}{3}$ cup (a measurement) and two $\frac{1}{3}$ -cup (a count) scoops are the same amount; one \$50 bill has the same value as fifty \$1 bills. However, in some circumstances the ambiguity leads to errors. One example is if I resolve to give the next beggar I encounter the smallest bill in my wallet, it matters if all I have is one \$50 or fifty \$1s. In the former case the value of my donation is much larger, even though the count of bills is the same. The significant difference is in the value, not the count. There are complicated but mundane reasons for how things acquire their value and why we can write "this bill is worth 5 USD", "the item on the shelf costs \$10", "silver is at \$80 per ounce" or "our team has 12 more points than the opposition", but these numbers aren't counts or measurements.

Suppose I am playing a simple basketball-like game: one where my score is the number of field goals I make within a given time, another which is similar but if I miss three times in a row I lose a point. In the former game the equivalence of the number of field goals and score is a rule and therefore an intentional part of play, a part I have removed in the latter game. This shows that score (value) and number of baskets (count) are distinct and separable, and in the second game I may coincidentally (or intentionally) have my score be equal to the number of field goals I make if I never incur a penalty.

Imagine a skeptic casting doubt on the fact $1+1=2$ by appealing to magnets, if you put the north ends of two bar magnets side-by-side they act together like a single magnet. We count: One magnet added to one magnet is one magnet, this apparently is due to the principle of superposition. But for mathematicians and calculators $1+1=2$ means that the *value* 1 plus the *value* 1 equals the *value* 2. These numbers are not derived from counting. An example where we confuse a count and a measurement is if I say, “in the 1960s the average household in the USA had 2.58 children” and someone takes this to mean that an average family has a 0.58 child, that 2.58 number is a measure of the average rather than a count of children.

We are also at risk of confusing values and measurements. What I can convince someone to pay for my home is not equivalent to the estimated value. Suppose there are four complete strangers in a room and between them they have a total of \$259.63 in their wallets; this sum is a dollar amount, so it looks like a value but isn't one. In the case where one of the strangers is carrying all the cash in the room the sum is both a measurement and a value. To use an example not involving money, suppose on a particular episode of a certain game show there are four contestants each having a score with a value in points, the average score is not a point-value, even though it's measured in points. If all four contestants have the same score, then the average will be equal to an actual point-value, furthering the confusion. All these examples show that we need a context to determine how the number is being used.

In contrast to this, the abacus is its own context. Numbers represented with an abacus are always treated as values which are determined by counting the active beads on

one or more adjacent rods. This disambiguation makes calculation more transparent than with paper arithmetic despite using the same numbers. Because we restrict ourselves to values when we are using an abacus, we aren't at risk of making a mistake like in the 0.58 child example. The abacus as a tool also provides a helpful reversal, counts and measurements are often concrete amounts of real-life things while values of points in a game and dollars are abstract, but when we move from written numbers to beads, we go from something more abstract, symbols, to something more concrete, counts of beads, thus we're able to reverse some of the abstraction required to do arithmetic. Ultimately the difference between the two ways of representing numbers is that written numbers require a context while the abacus is a context, and it can teach its user to better understand arithmetic which is a context where numbers are always values.